

MARKETING CAMPAIGN ANALYSIS

using A/B Testing & Regression Analysis

PROJECT REPORT

Python | Pandas | NumPy | Matplotlib | Seaborn
SciPy | Scikit-learn | Statsmodels

Facebook Ads vs AdWords Ads

2019 Daily Campaign Data — 365 Records

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Data Analytics Portfolio Project

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1. Project Overview

This project analyzes daily digital advertising campaign data for two platforms: Facebook Ads and AdWords. The analysis covers the full year 2019 and focuses on campaign activity, clicks, conversions and advertising cost.

The purpose is to compare the two campaigns using both descriptive analysis and statistical methods. The work starts with data understanding and exploratory data analysis, then moves into campaign comparison, correlation analysis, A/B testing through hypothesis testing, Linear Regression, weekly and monthly analysis, cost per conversion and a cointegration test.

The report follows the same practical sequence used in the notebook so that each result can be traced back to a part of the analysis.

2. Business Problem

A marketing agency is running advertising campaigns on Facebook and AdWords. The agency wants to understand how the two platforms perform and whether the difference in their conversion results is large enough to be statistically meaningful.

The analysis looks at clicks, conversions, campaign efficiency and changes in performance over time. The goal is to turn the daily campaign records into information that can be understood by a marketing or business team.

3. Research Questions

- How does Facebook campaign performance compare with AdWords?
- Do more clicks tend to be associated with more conversions?
- Is the difference in conversion performance statistically significant?
- Can Facebook conversions be estimated from Facebook ad clicks?
- How do Facebook conversions change across the year and across weekdays?
- How does cost per conversion change from month to month?
- Is there evidence of a long-term relationship between Facebook advertising cost and conversions?

4. Dataset and Variables

The dataset contains 365 daily records and 17 columns. The date range is January 1, 2019 to December 31, 2019. Each record represents one day's campaign activity.

Main Facebook variables

- Facebook Ad Views
- Facebook Ad Clicks
- Facebook Ad Conversions
- Cost per Facebook Ad
- Facebook Click-Through Rate
- Facebook Conversion Rate
- Facebook Cost per Click

Main AdWords variables

- AdWords Ad Views
- AdWords Ad Clicks

- AdWords Ad Conversions
- Cost per AdWords Ad
- AdWords Click-Through Rate
- AdWords Conversion Rate
- AdWords Cost per Click

The Date field is used to create time-based views such as month and weekday.

5. Tools and Libraries

- Python — complete analysis
- Pandas — data loading, cleaning and manipulation
- NumPy — numerical operations
- Matplotlib and Seaborn — charts and visual analysis
- SciPy — statistical testing
- Scikit-learn — Linear Regression and evaluation
- Statsmodels — cointegration analysis

6. Analysis Workflow

Business Problem → Research Questions → Data Understanding → Data Preparation → EDA → Campaign Comparison → Correlation Analysis → A/B Testing / Hypothesis Testing → Regression Analysis → Weekly and Monthly Analysis → Cost Per Conversion → Cointegration Analysis → Business Findings

7. Data Loading and Preparation

The CSV file is loaded with Pandas and the first rows, shape, data types and descriptive statistics are checked. The Date column is converted to datetime so that time-based analysis can be performed.

The source data contains dates in slash-separated formats, so the notebook uses mixed-format parsing when converting the Date column. This keeps the original data values intact while making the column usable for grouping and time analysis.

The preparation stage is intentionally simple because the dataset is already structured for analysis. The main preparation work is understanding the columns, converting the date and preparing the fields needed for later analysis.

8. Exploratory Data Analysis

EDA is used to understand the campaign data before statistical testing. The notebook examines the distributions of clicks and conversions and uses visualizations to compare the behavior of the two campaigns.

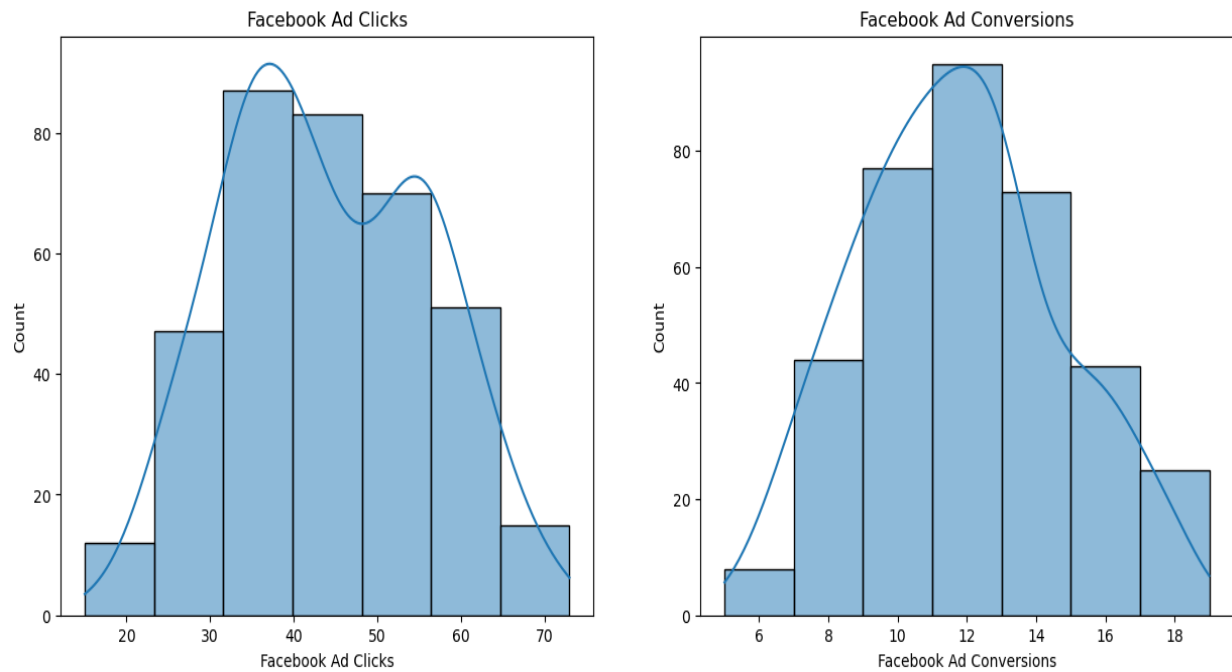


Figure 1. Distribution of clicks and conversions

The distribution charts provide a quick view of the typical range of daily clicks and conversions and help identify whether the variables are concentrated or spread across a wider range.

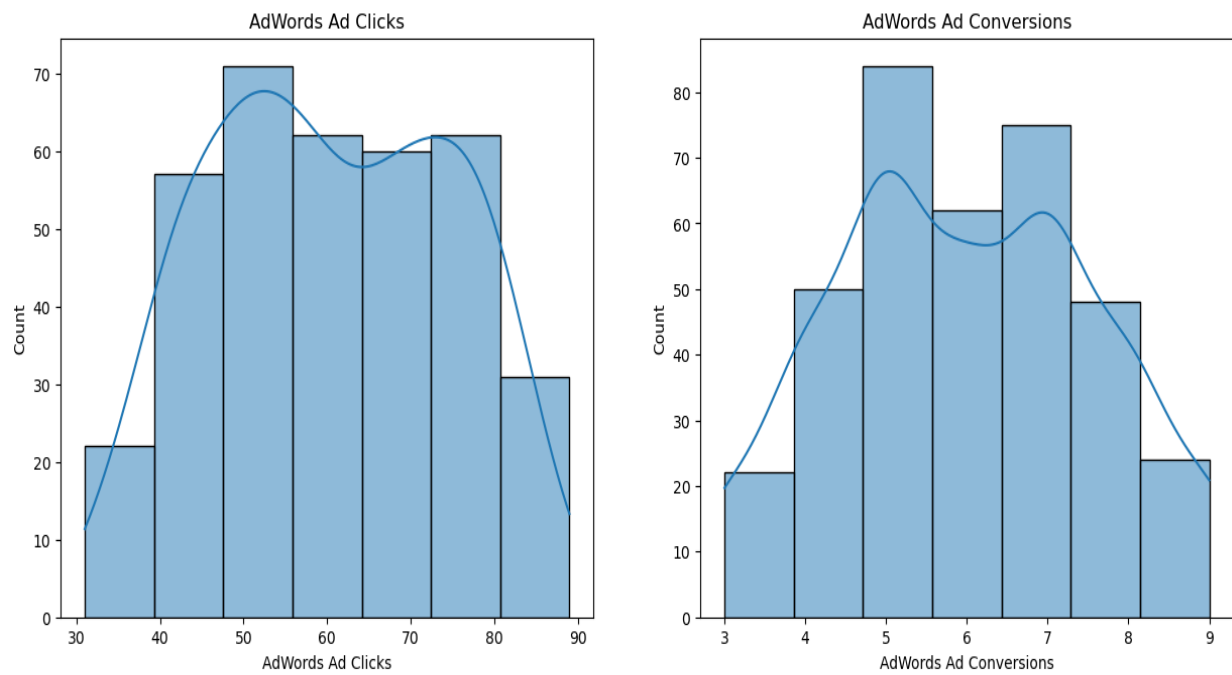


Figure 2. Additional distribution view from the campaign analysis

The visual checks are followed by conversion-category analysis and correlation analysis. The purpose is to understand the data before drawing statistical conclusions.

9. Campaign Performance Comparison

Daily conversions are grouped into four categories: less than 6, 6–10, 10–15 and more than 15 conversions.

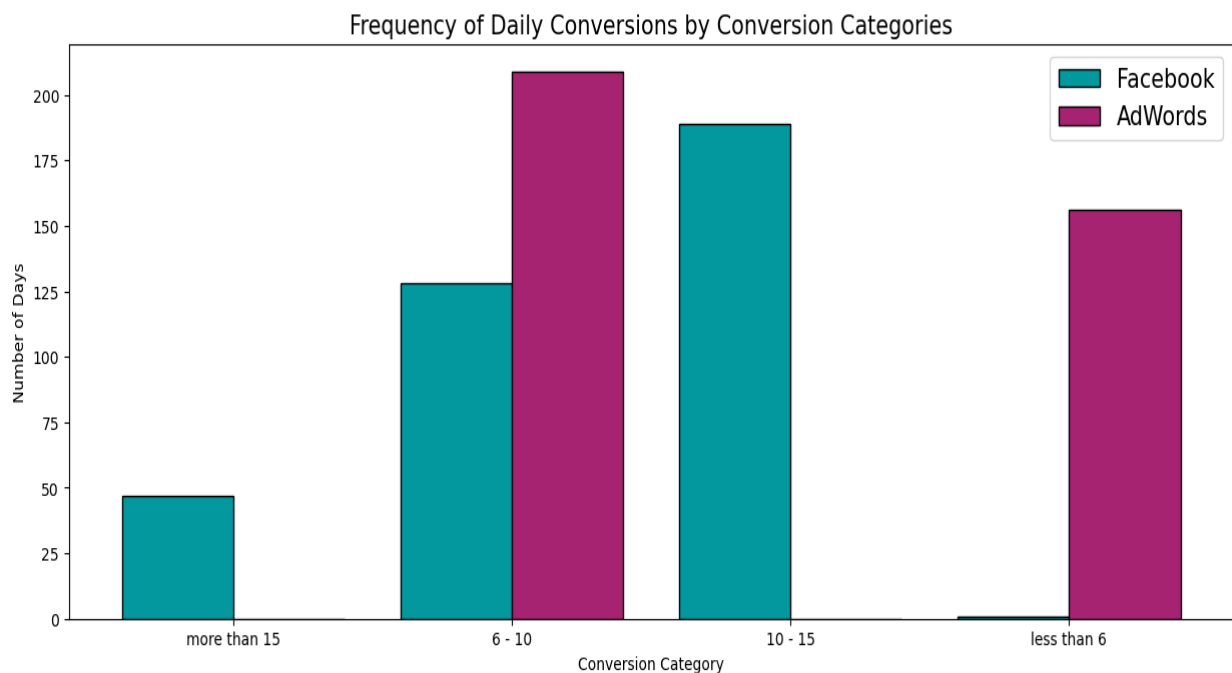


Figure 3. Facebook and AdWords conversion categories

The category comparison shows that Facebook has more frequent higher-conversion days, while AdWords has a greater concentration in lower conversion ranges.

10. Clicks and Conversions

Scatter plots are used to examine whether higher click activity is associated with higher conversion activity for each platform.

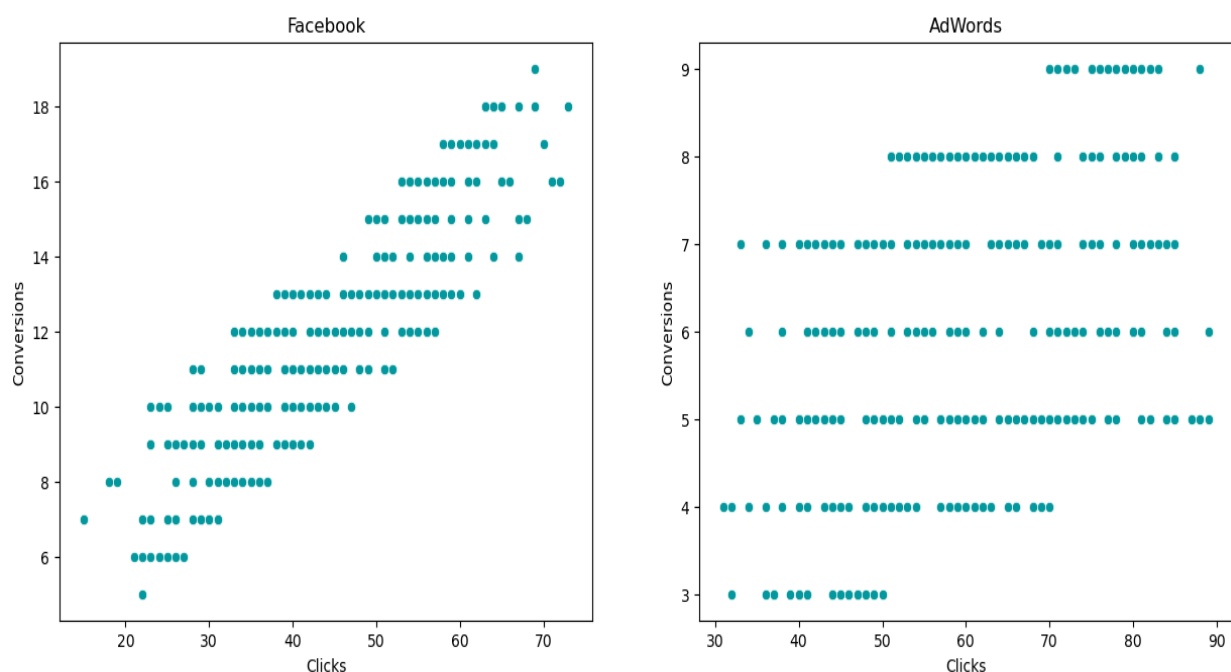


Figure 4. Clicks versus conversions for Facebook and AdWords

The Facebook click-to-conversion relationship is visibly stronger than the corresponding AdWords relationship. The notebook reports a correlation of approximately 0.87 for Facebook and approximately 0.45 for AdWords. Correlation describes association and does not by itself establish causation.

11. A/B Testing and Hypothesis Testing

For the campaign comparison, Facebook is treated as group A and AdWords as group B. The A/B testing idea is the comparison of the two campaign groups, while hypothesis testing provides the statistical method for deciding whether the observed difference is significant.

11.1 Hypotheses

Null hypothesis (H_0): Facebook does not have higher average conversions than AdWords.

Alternative hypothesis (H_1): Facebook has higher average conversions than AdWords.

The notebook uses a 5% significance level and performs an independent-samples statistical test on the conversion values.

11.2 Results

Measure	Result
Mean Facebook conversions	11.74
Mean AdWords conversions	5.98
T-statistic	Approximately 32.88
P-value	Approximately 9.35×10^{-13}
Significance level	0.05

The p-value is far below 0.05. Based on the test used in the notebook, the null hypothesis is rejected. The result provides strong statistical evidence of a difference in average conversion performance, with Facebook having the higher sample mean in this dataset.

12. Regression Analysis

Regression is used to examine whether Facebook ad clicks can help estimate Facebook ad conversions. The independent variable is Facebook Ad Clicks and the dependent variable is Facebook Ad Conversions.

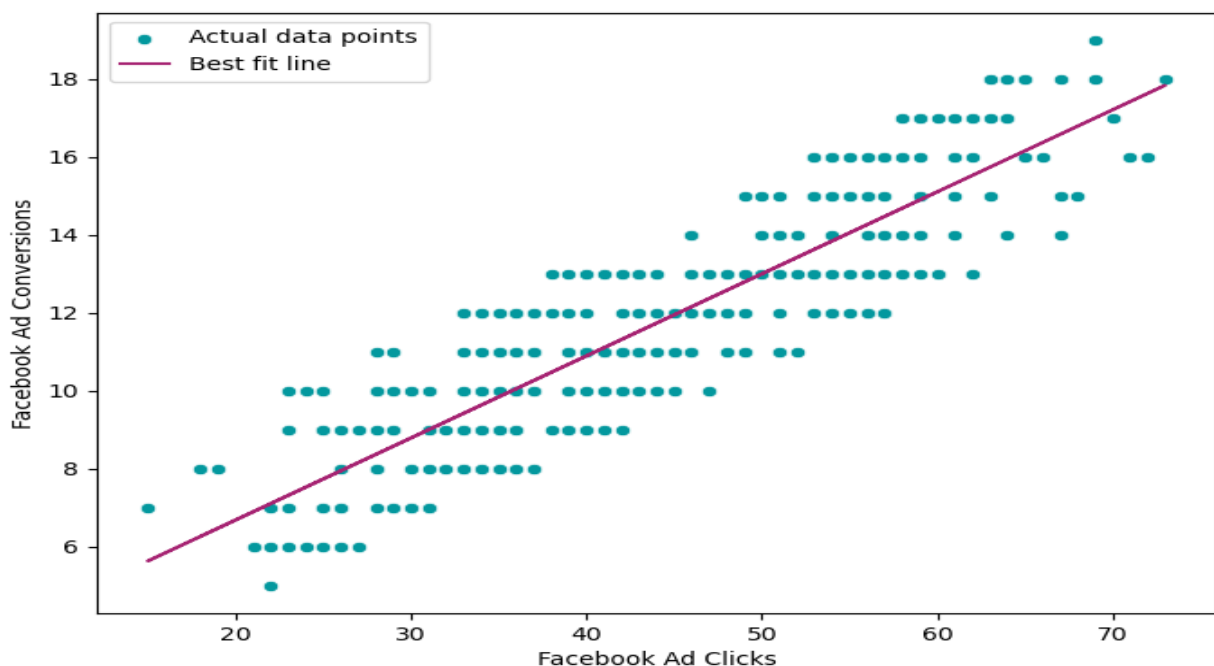


Figure 5. Linear Regression: Facebook clicks and conversions

The upgraded regression workflow uses an 80/20 train-test split. The model is fitted on training data and evaluated on unseen test data using R^2 Score and Mean Squared Error.

The earlier notebook version fitted and evaluated the model on the same observations and reported an R^2 of approximately 76.35%. That number describes the fitted sample, not unseen test performance. The upgraded version avoids this issue by evaluating predictions on the held-out test set.

The original analysis also demonstrates predictions for specific click levels, including 50 and 80 clicks, to show how a regression equation can be translated into an expected conversion value.

13. Facebook Campaign Analysis Over Time

The analysis focuses on Facebook and extracts month and weekday information from the Date column.

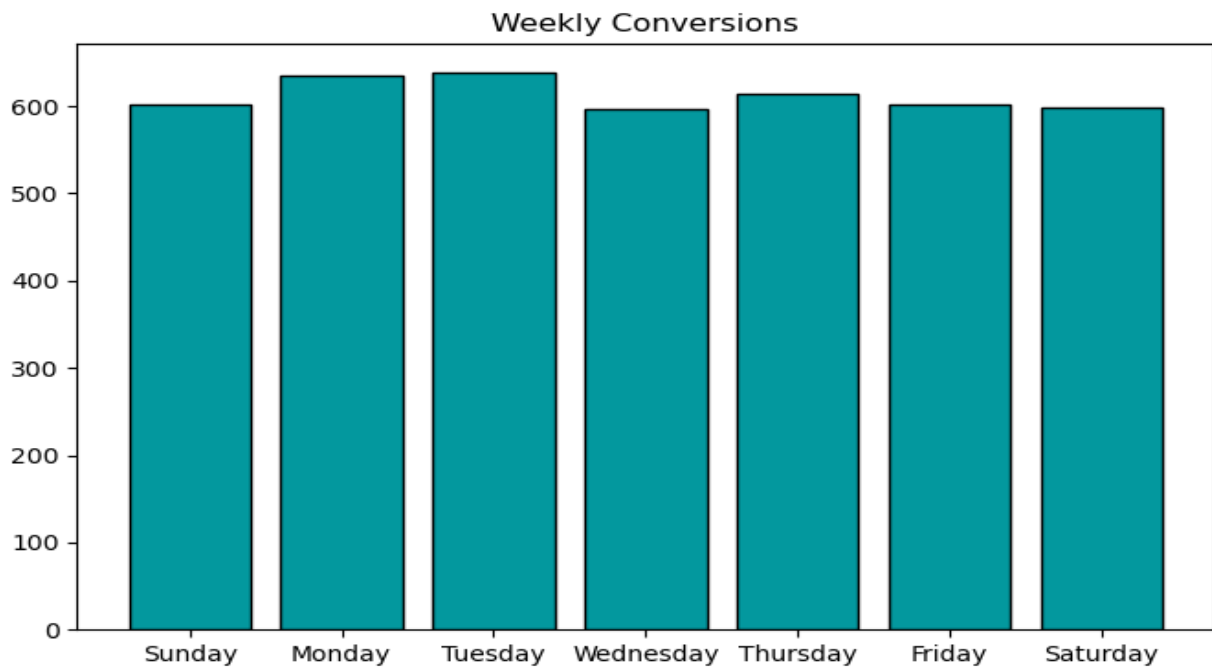


Figure 6. Weekly conversion analysis

The weekday view shows that total conversions remain relatively consistent across the week, with some differences between individual weekdays.

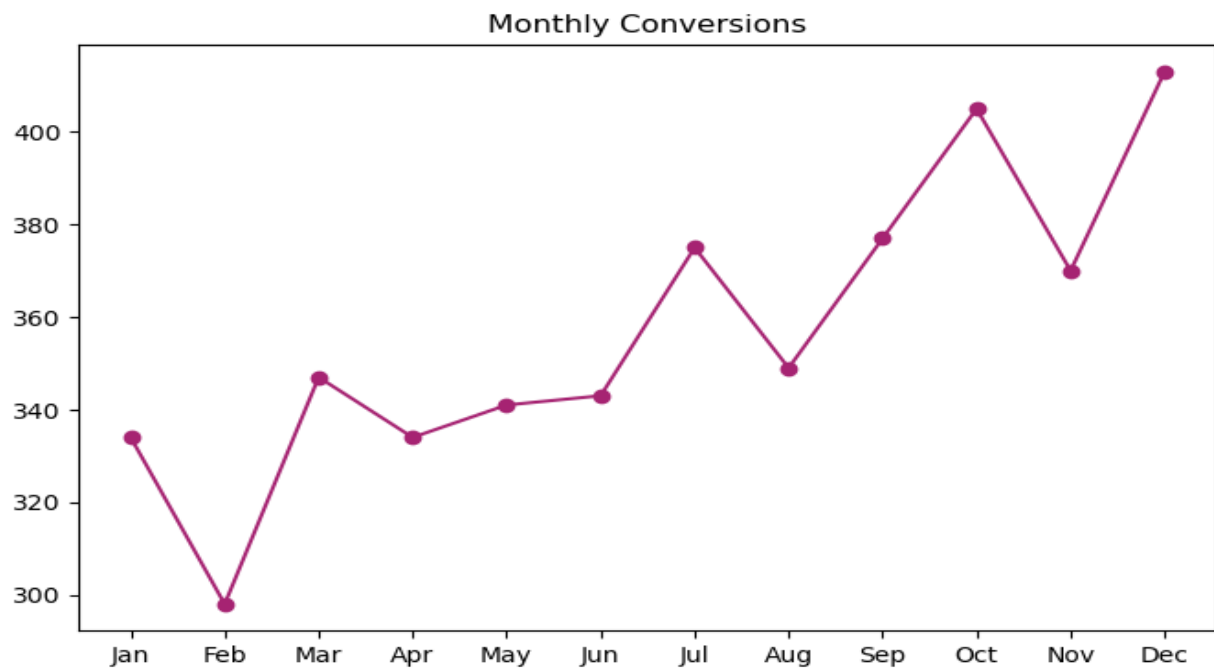


Figure 7. Monthly conversion analysis

The monthly view shows changes in conversion volume during the year. Looking at each month separately makes it easier to identify stronger and weaker periods that are hidden by a single yearly average.

14. Cost Per Conversion

Monthly cost per conversion is calculated as total Facebook advertising cost divided by total Facebook conversions for that month.

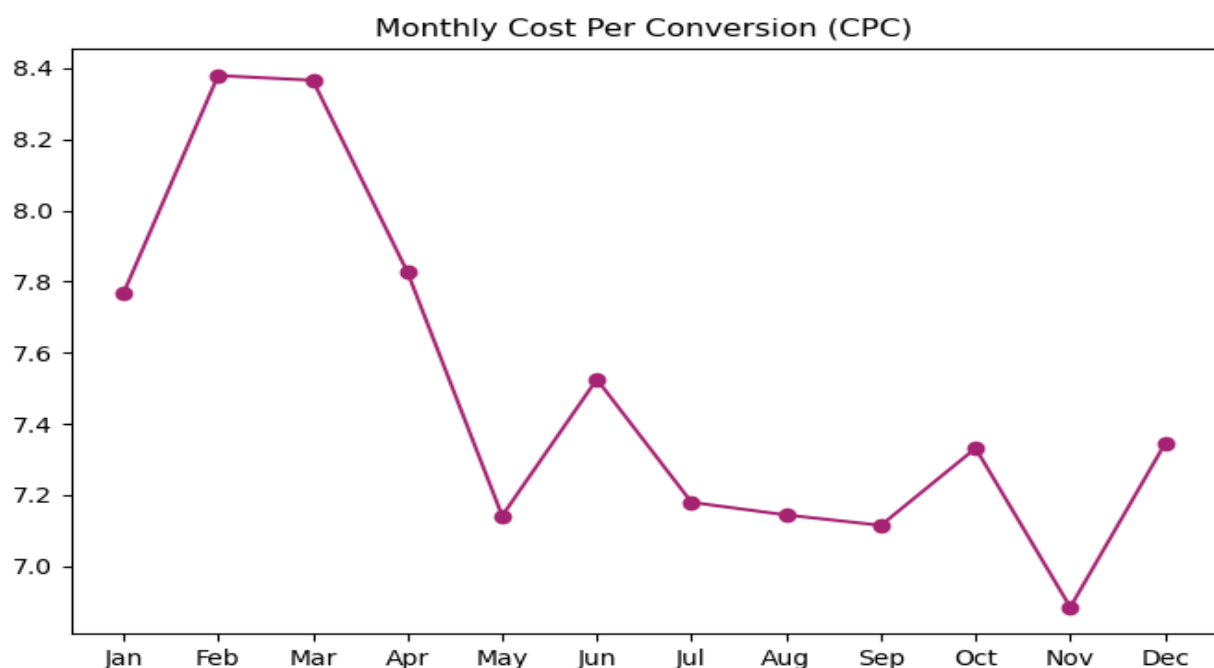


Figure 8. Monthly cost per conversion

The notebook reports fluctuations in monthly cost per conversion. May and November are identified as relatively lower-cost months, while February is identified as a relatively higher-cost month.

15. Cointegration Analysis

The final statistical section examines whether Facebook advertising cost and Facebook conversions show evidence of a long-term equilibrium relationship.

A cointegration test is used for this purpose. The notebook reports a p-value below the selected 0.05 significance level and rejects the null hypothesis of no cointegration. This provides evidence of a long-term relationship between the two time series in the analyzed data.

This result is different from a simple correlation result because the cointegration test is concerned with long-term movement between time series.

16. Key Results

Analysis	Result
Dataset	365 daily records, 17 columns
Period	January 1, 2019 – December 31, 2019
Average Facebook conversions	11.74
Average AdWords conversions	5.98
Facebook clicks vs conversions	Correlation ≈ 0.87
AdWords clicks vs conversions	Correlation ≈ 0.45
A/B test t-statistic	≈ 32.88
A/B test p-value	$\approx 9.35 \times 10^{-13}$
Regression predictor	Facebook Ad Clicks
Regression target	Facebook Ad Conversions
Earlier same-data R^2	$\approx 76.35\%$
Lower-cost months noted	May and November
Higher-cost month noted	February

17. Business Findings

- Facebook has a higher average conversion count than AdWords in the analyzed 2019 data.
- The hypothesis test provides strong statistical evidence of a difference in the average conversion values.
- Facebook clicks have a much stronger positive correlation with Facebook conversions than AdWords clicks have with AdWords conversions in this dataset.
- The regression analysis shows how click activity can be used to estimate Facebook conversions.
- Conversion totals change across weekdays and months, so campaign performance is not constant throughout the year.
- Cost per conversion also changes from month to month, with May and November identified as relatively lower-cost periods and February as a relatively higher-cost period.
- The cointegration result provides evidence of a long-term relationship between Facebook advertising cost and Facebook conversions in the analyzed series.

18. Overall Conclusion

This project combines practical marketing analysis with statistical techniques. It begins with EDA to understand the daily campaign data, compares Facebook and AdWords conversion behavior, tests the difference statistically, models Facebook clicks and conversions through Linear Regression, and then studies campaign performance over time.

Within the available 2019 data, Facebook records the higher average conversion count. The hypothesis test supports a statistically significant difference in the sample, while the correlation and regression analysis show that Facebook click activity is strongly associated with Facebook conversions.

The time-based analysis adds another layer by showing that conversions and cost per conversion change across the year. The cointegration analysis then examines the longer-term relationship between advertising cost and conversions.

Overall, the project demonstrates a Python-based workflow for turning daily marketing campaign records into statistical and business insights.